How-To Library

Informatica

Importing Kafka Topics in Binary Format in PowerCenter

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Abstract

You can use PowerExchange for Kafka to import Kafka topics as a source or target definition in the PowerCenter Designer. This article describes how to import Kafka topics as source or target definitions in binary format.

Supported Versions

- PowerExchange for Kafka 10.4.0

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Overview

Apache Kafka is a publish-subscribe messaging system that manages and maintains the real-time stream of data. With Kafka, any user or application can write any messages to any topic or read data from any topic. You can use the PowerCenter Designer to import Kafka topics as source or target definitions.

Verify Prerequisites

Before you import a Kafka topic in Binary format, complete the following tasks:

1. Install or upgrade PowerCenter services and PowerCenter Client.
2. Ensure that you register the PowerExchange for Kafka plug-in with the PowerCenter repository.
3. Ensure that you install the Apache Kafka cluster and the Informatica Services in different nodes.

For more information about product requirements and supported platforms, see the Product Availability Matrix on Informatica Network:
https://network.informatica.com/community/informatica-network/product-availability-matrices.

Kafka Sources and Targets Overview

You can create a mapping with a Kafka source to read data from a Kafka topic. You can create a mapping with any source and a Kafka target to write data to a Kafka topic.

A Kafka source or target represents data in a Kafka stream. Create a PowerExchange for Kafka source or target to read from or write to Apache Kafka brokers.

Kafka runs as a cluster comprised of one or more servers each of which is called a broker. Kafka brokers stream data in the form of messages. These messages are published to a topic.

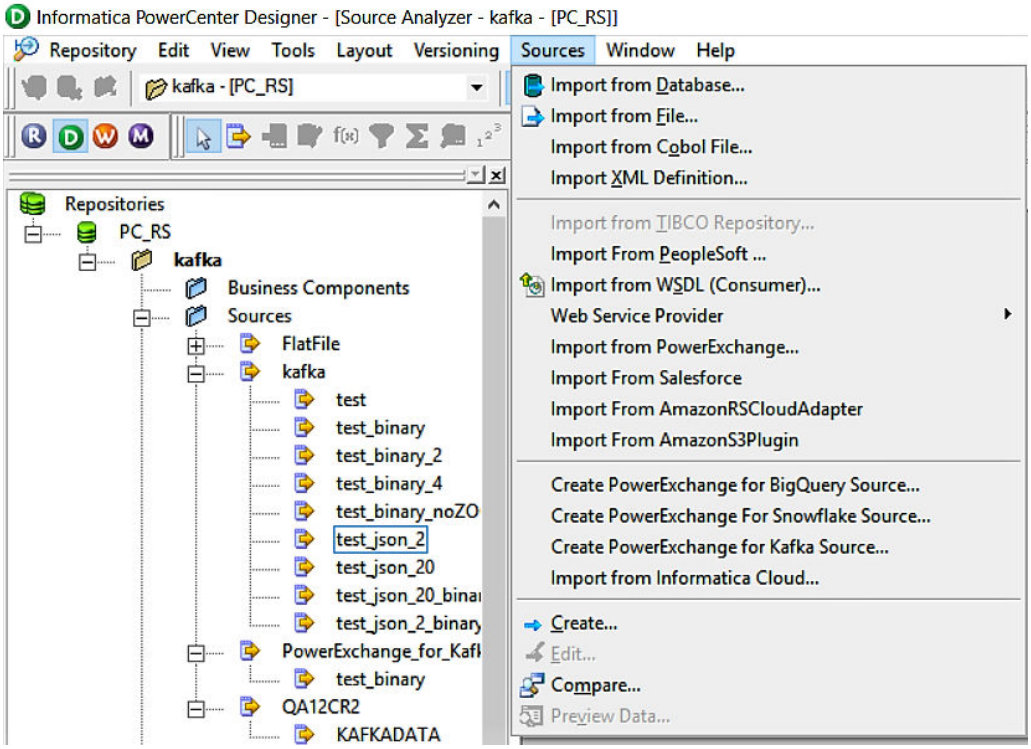
When you import a Kafka source or target, select the topic name that you want to read from or write to. You can specify the topic name or use a regular expression for the topic name pattern only when you read from Kafka. To subscribe to multiple topics that match a pattern, you can specify a regular expression. When you run the application on the cluster, the pattern matching is done against topics before the application runs. If you add a topic with a similar pattern when the application is already running, the application will not read from the topic.

Import Kafka Source and Target Definitions in Binary Format

Use the **Create PowerExchange for Kafka Source** or **Create PowerExchange for Kafka Target** wizard to import Kafka source and target definitions into the PowerCenter repository.

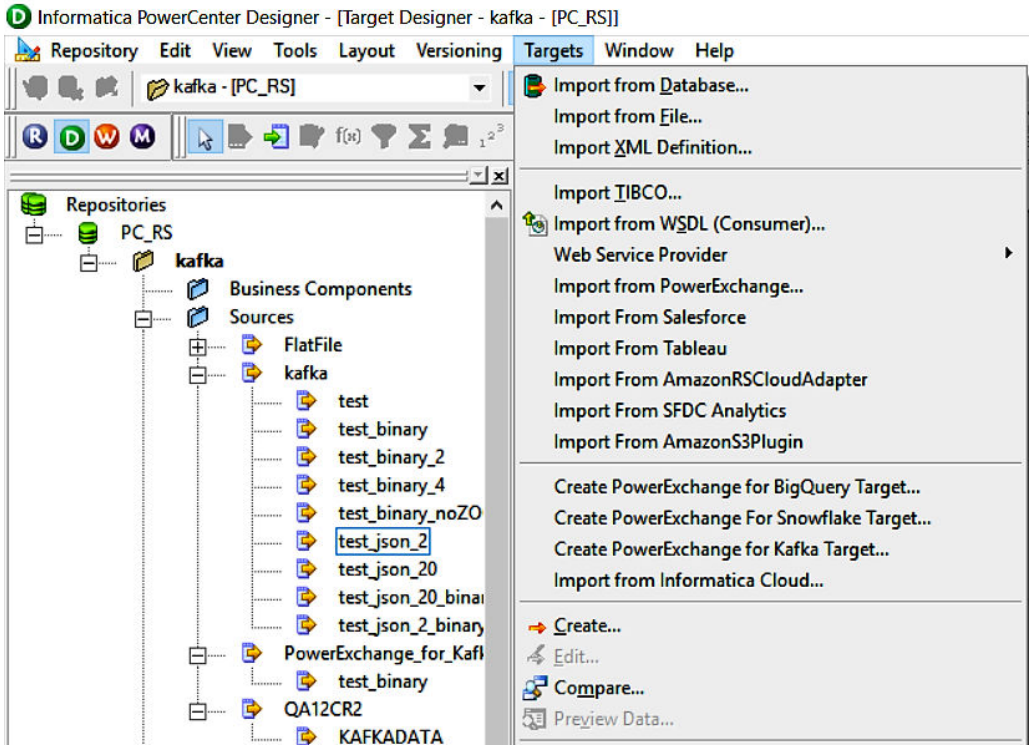
You must import Kafka source and target objects before you create a mapping.

1. Start PowerCenter Designer, and connect to a PowerCenter repository configured with a Kafka instance.
2. Open a source or target folder.
3. Select **Source Analyzer** or **Target Designer**.
   - In the Source Analyzer, click **Sources** > **Create PowerExchange for Kafka Source** as shown in the following image:



     The **Kafka Connection** wizard appears.

- In the Target Analyzer, click **Targets** > **Create PowerExchange for Kafka Target** as shown in the following image:



  The **Kafka Connection** dialog box appears.

4. Configure the following connection parameters:

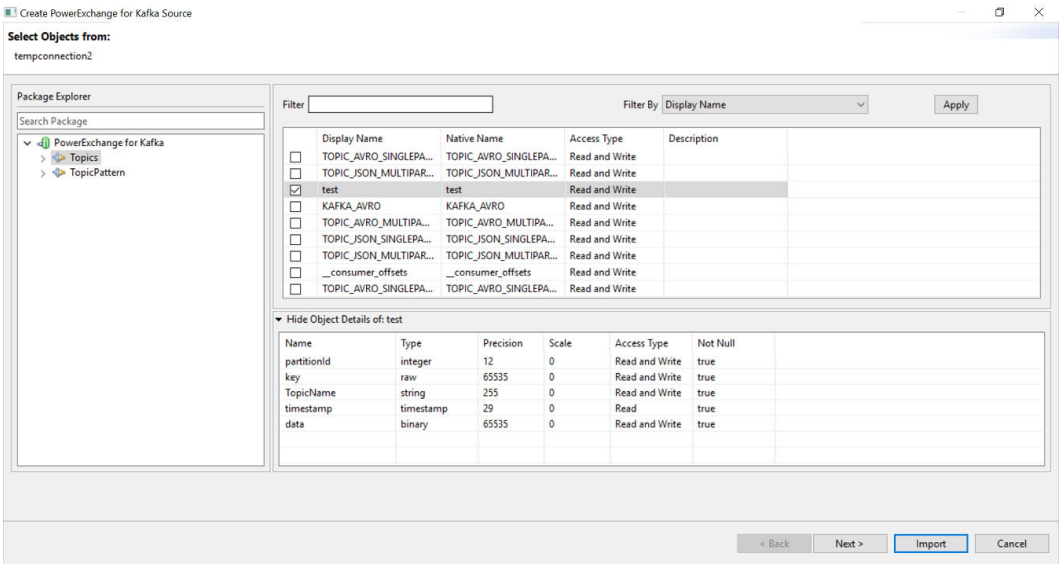
Connection Parameter	Description
Kafka Broker List	The IP address and port combinations of the Kafka messaging system broker list. The IP address and port combination has the following format: `<IP Address>:<port>` You can enter multiple comma-separated IP address and port combinations.
Retry Timeout in seconds	Number of seconds the Integration Service attempts to reconnect to the Kafka broker to write data. If the source or target is not available for the time you specify, the mapping execution stops to avoid any data loss. Default is 180 seconds.
Kafka Broker Version	Select **Apache 0.10.1.1 & above** as the Kafka messaging broker version.

Connection Parameter	Description
Additional Connection Properties	Comma-separated list of connection properties to connect to the Kafka broker. To connect to a Kafka broker over SSL, ensure that you set the following properties: - security.protocol - ssl.truststore.location - ssl.truststore.password If the `ssl.client.auth` parameter is set to `required` or `requested`, ensure that you set the following properties: - ssl.keystore.location - ssl.keystore.password - ssl.key.password For example, `security.protocol=SSL,ssl.truststore.location=<complete path and name of the client.truststore.jks file>,ssl.truststore.password=<client truststore password>,ssl.keystore.location=<complete path and name of the client.keystore.jks file>,ssl.keystore.password=<client keystore password>,ssl.key.password=<password for the individual key>` You can also specify the configuration properties required to connect to a Kerberised Kafka cluster. Specify the following property: `security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=GSSAPI`
SSL Mode	Specifies whether the PowerCenter Integration Service establishes a secure connection to the Kafka broker.You can select one of the following options: - disabled. The PowerCenter Integration Service establishes an unencrypted connection to the Kafka broker. - One-Way. The PowerCenter Integration Service establishes an encrypted connection to the Kafka broker using truststore file and truststore password. - Two-Way. The PowerCenter Integration Service establishes an encrypted connection to the Kafka broker using truststore file and truststore password.
SSL TrustStore File Path	Applicable only if you select **One-Way** or **Two-Way** as the SSL mode. The complete path and file name of the truststore file. The truststore file contains the SSL certificate that the Kafka cluster validates against the Kafka broker certificate.
SSL TrustStore Password	Applicable only if you select **One-Way** or **Two-Way** as the SSL mode. The password for the truststore file.
SSL KeyStore File Path	Applicable only if you select **Two-Way** as the SSL mode. The complete path and name of the Java keystore file. The keystore file contains the certificate that the Kafka broker validates against the Kafka cluster certificate
SSL KeyStore Password	Applicable only if you select **Two-Way** as the SSL mode. The password for the keystore file.
Additional Security Properties	Not applicable for PowerExchange for Kafka.

5. Click **Test** to test the connection.
6. Click **Finish** to add the connection.

   The **Select Objects from** tab appears.
7. Select the **Topics** package in **Package Explorer**.

A list of topic appears as shown in the following image:



8. To view the topic metadata, select the topic, and double-click the topic name.

   Note: You cannot select multiple topics in a Kafka source or target definition.

9. Select the topic that you want to import, and then click **Import**.

   Note: You cannot select topics from the **Topics Pattern** package.

   PowerExchange for Kafka imports the data from the Kafka topic in binary format.

Author

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